

Alert Investigation Report

Platform: SOC Simulation Project — WDLabs

Environment: Microsoft Defender for Endpoint / Microsoft Sentinel

Case Details

Field	Value
Alert Name	## WDigest configuration change
Alert ID / Link	https://security.microsoft.com/alerts/da12ff5b6e-e4f8-4443-8bd8-9c6d550e1f92_1?tid=567ea528-4e82-40fa-914e-38c05fd69677
Analyst	Carl Samson
Date Investigated	22/09/26
Severity	Low

1. Alert Summary

- Host / Account affected:** socsim-srv01 / Administrator
- Date & Time of alert:** 25/8/2026 20:50:29
- What triggered the alert:** WDigest changes runned by powershell commands
- Why it triggered (rule/logic):** Behaviour alerts

2. Investigation Findings

What happened?

The activity started with an elevated `cmd.exe` process (PID 2872) at 20:50:26, which spawned `powershell.exe` a second later. That first PowerShell process (PID 468) was launched with `-NoProfile -NonInteractive -ExecutionPolicy Unrestricted -EncodedCommand`, passing its actual instructions as a base64 blob rather than plain text on the command line. Decoding that blob (base64 to UTF-16LE, which is how PowerShell always encodes `-EncodedCommand`

payloads) shows it wasn't the real action at all, it was PowerShell calling itself again with a second, nested `-EncodedCommand`, which is what spawned the child process PID 4196.

Decoding that second, inner blob is where the actual executed code shows up: a script that sets the console's input/output encoding to UTF-8, checks that the PowerShell version is at least 3.0, reads a payload piped in over stdin, splits that payload on a four-null-character delimiter into a script block and a JSON argument block, then runs the script block with

`[ScriptBlock]::Create(...)`. That exact script is the standard Ansible Windows/WinRM "exec wrapper", the mechanism Ansible's `win_*` modules use to push and execute code on a remote Windows host over WinRM. So the trigger for this whole chain wasn't a person typing commands interactively; it was remote command execution delivered through Ansible automation, running under the `SOCSIM-SRV01\Administrator` account with high integrity but standard (non-elevated-token) execution.

Two seconds after that wrapper ran, the same PowerShell process (PID 4196) made the one change that actually fired the alert: it wrote to

`HKLM\SYSTEM\ControlSet001\Control\SecurityProviders\WDigest`, setting the value `UseLogonCredential` from empty to `1`. That registry write is what the alert is built around — the double-encoded PowerShell and the Ansible wrapper were just the delivery path that got a script running on the box; the WDigest change is the actual technique being executed, and it's what forces Windows to start caching the next logon's password in plaintext in LSASS memory instead of only a hashed form.

WDigest Alert — Process Flow

Host SOCSIM-SRV01 · 25 Aug 2026, 20:50:26–29 · User Administrator



● Elevated shell / process start ● Encoded command layer ● Decoded — legitimate automation code ● Alert-triggering action

Evidence

// Paste key log entries, process trees, or notable findings here

Decoded powershell command

Process tree

Alert timeline

Expand all

Copy story to clipboard

25/8/2026

20:50:29

WDigest configuration change

Low

Detected

Resolved (Informational, expected activity)

25/8/2026

20:50:29

[4196] powershell.exe changed registry value SYSTEM\ControlSet001\Control\SecurityProviders\WDigest

Key hiveHKEY_LOCAL_MACHINE

Value nameUseLgcnCredential

Set value data1

Original value data<empty>

Initiating process

[4196] powershell.exe -NoProfile -NonInteractive -ExecutionPolicy Unrestricted -EncodedCommand dABYAHkAIA87ACAAWwBDAG8Abg8zAG8AbAIAF0A0gA6EAkAbgBwAHUADABFAG4...

Command line

"powershell.exe" -NoProfile -NonInteractive -ExecutionPolicy Unrestricted -EncodedCommand dABYAHkAIA87ACAAWwBDAG8Abg8zAG8AbAIAF0A0gA6EAkAbgBwAHUADABFAG4AYwBVAGQAA0BuAgCATAIA9ACAAWwBDAG8Abg8zAG8AbAIAF0A0gA6EA8dQB0HAAdQB0AEIuBgbJA8GAZ1pGAC4AZwAgAD0A1ABDIAUAD0A1AE5AYgBgAGUAYwB8ACAAUwBSAHMADABALgBAGUAGUAE8BACAAVQBUEVADABGAGAYwBVvAGQAA0BuAgCATAB0ACAYwBHAHQAYwB8ACAAwAGCAG0AB3IAUAD0A1AAFA87ABBA0A0g0BmCAAKAAFAAF0UwBwAGUAcgBzAGkABwBIAF0A0g1AGCwZ0A0FAAFuBwBAGCgBzAGkABwBuAACB0ABQAB08A0HQA1ABBAFAFVAYZ0BvAHMA0gBvAG4AQXA1ADMDgAwACTAK0AgAHSACgAnAht1qBmGEA0g0B8AGUZA1AD0AdABwAHUZA20ASACTAB0BZAGCAtIqA6ACTIAQ0BUAHMA0QB1AGwAZ0AQHIAZQBxAHUAA0QYbAGUAcwAgAFABwB3AGUAcgBTAAG0AZQB8AGwAIAB2ADMDgAwACAAbwByACAABgBIAHCAZQBvACTAT0AnAA0AZQB4AGkAdAAGADEACgB9AA0AJAB1AH0AZ0B1AF8ADwBybGEACeABwAGUAcgBfAHMAdABYcACAP0AgAC0A0BuAHAA0dQB0ACAAFAAg0AE8dQB0AC0AluB0AHIAA0BuAGACgAKAHMAcABsAGkAdABFAHAAYQYbAHQACwAgAD0A1AAkAGUAEAB1AGMAxwB3AHIAAY0BwAHAAZ0YbAF8ACwB0AHIALgBTAHAABABpAHQAKABAAcAgIgbGADAAyAAwAGMAABgdADA1tAgpACwAIAAyaCwAt1ABbAFMADByAGkABgBnAFMAcABsAGkAdABPAHAADABpAgB8AGBzAF8A0GAF6AFIAZQBTA8GdABgB1AEUABQBwAHQAEQBFAG4AdABYAGMAZ0CkACkABgBJAGY1AA0AC0ABgBwAHQA1AAkAHMAcABsAGkAdABFAHAAYQYbAHQACwAUEwAZQBwAGUAcgAdAB0AACAL0B1AHEAIEAAYcACKA1AB7ACAAADBoAHTABwB3CAAIgBpAG4AGdBgBAGwAQkACAACABhAHkABABvAGEAZA1ACAA70AKAFMAZQB0AC0Ag0BvAHhIAA0QbHAGIAbAB1ACAA1QB0AGAEAb0B1ACAAAg8zAG8AbgBfFAHIAY0B3ACAA1QBWAGAEAbA1AGUAI1AAkAHMAcABsAGkAdABFAHAAYQYbAHQACwBbADEAXQAKACy1AA0AFsAUwBjAHIAA0QBwAHQAA0gB8AGBYwRfAF0A0gA6EAEMcgb1AGEAdAB1ACgAJABzAHAABABpAHQAXwBwAGEAgCB0AHMAWwAwF8AKQAPAA==

Command line (decoded)

try { [Console]::InputEncoding = [Console]::OutputEncoding = New-Object System.Text.UTF8Encoding } catch { \$null = \$_
if (\$PSVersionTable.PSVersion -lt [Version]"3.0") {
'{"failed":true,"msg":"Ansible requires PowerShell v3.0 or newer"}'
exit 1
}
\$exec_wrapper_str = \$input | Out-String
\$split_parts = \$exec_wrapper_str.Split(@(('0'0'0'0')), 2, [StringSplitOptions]::RemoveEmptyEntries)
If (-not \$split_parts.Length -eq 2) { throw "invalid payload" }
Set-Variable -Name json_raw -Value \$split_parts[1]
& ([ScriptBlock]::Create(\$split_parts[0]))

Process tree timeline

Process tree

Alert timeline

Copy to clipboard

Expand all Copy story to clipboard

25/8/2026 20:50:26 [2872] cmd.exe

20:50:27 [468] powershell.exe PowerShell -NoProfile -NonInteractive -ExecutionPolicy Unrestricted -EncodedCommand UA...

20:50:27 [4196] powershell.exe -NoProfile -NonInteractive -ExecutionPolicy Unrestricted -EncodedCommand dABYAHk...

20:50:29 [4196] powershell.exe changed registry value SYSTEM\ControlSet001\Control\SecurityProviders\WDigest

WDigest configuration change

Low (Informational, expected activity)

KQL commands on finding other commands running outside the alert timeline

Microsoft Defender

Search

Advanced hunting

Selected workspace: wdlabs

Help resources

Query resources report

Explore your content from Microsoft Sentinel

All data from Microsoft Sentinel, including tables, queries and functions is now available for you to explore.

Close

New query* x New query* x New query* x

Schema

Alerts & behaviors

AlertEvidence

AlertInfo

BehaviorEntities

BehaviorInfo

Apps & identities

IdentityAccountInfo

IdentityInfo

Email & collaboration

Devices

DeviceEvents

Timestamp

DeviceId

DeviceName

ActionType

FileName

FolderPath

Query

Run query

1 DeviceProcessEvents

2 | where DeviceName has "SOC5IM-SRV01"

3 | where ProcessId == 2872

4 | project Timestamp, FileName, ProcessCommandLine, InitiatingProcessFileName, InitiatingProcessCommandLine, AccountName

Results

Query history

Getting started

Export

Show empty columns

18 items

Search

00:01:515

Low

Timestamp

FileName

ProcessCommandL...

InitiatingProcessFil...

InitiatingProcessCo...

AccountName

19 Sep 2026 04:31...

powershell.exe

powershell.exe "Import-...

gc_arc_service.exe

"gc_arc_service.exe" -k n...

system

19 Sep 2026 05:25...

TrustedInstaller.exe

TrustedInstaller.exe

services.exe

services.exe

system

19 Sep 2026 12:04...

auditpol.exe

"auditpol.exe" /get /sub...

powershell.exe

powershell.exe -Executio...

system

20 Sep 2026 22:04...

powershell.exe

powershell.exe -Executio...

senselr.exe

"SenselR.exe" "OfflineSe...

system

8 Sep 2026 23:38:49

ngen.exe

"ngen.exe" RemoveTask...

ngentask.exe

"NGenTask.exe" /Runtim...

system

13 Sep 2026 23:38...

svchost.exe

svchost.exe -k NetworkS...

services.exe

services.exe

network service

Export

Show empty columns

9 items

Search

00:01:688

Low

Filters

Add filter

Timestamp

FileName

ProcessCommandLine

InitiatingProcessFil...

AccountName

8 Sep 2026 07:36:21

powershell.exe

powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command "& {if (\$ExecutionContext.SessionSt...

senselr.exe

system

8 Sep 2026 20:29:20

conhost.exe

conhost.exe 0xffffffff -ForceV1

sc.exe

himds

12 Sep 2026 20:28...

conhost.exe

conhost.exe 0xffffffff -ForceV1

sc.exe

himds

12 Sep 2026 22:32...

conhost.exe

conhost.exe 0xffffffff -ForceV1

powershell.exe

system

13 Sep 2026 08:42...

cvtres.exe

cvtres.exe /NOLOGO /READONLY /MACHINE:IX86 "/OUT:C:\Windows\SystemTemp\RES3E87.tmp" "c:\Windows\Syste...

csc.exe

system

15 Sep 2026 11:52...

cvtres.exe

cvtres.exe /NOLOGO /READONLY /MACHINE:IX86 "/OUT:C:\Windows\SystemTemp\RES690D.tmp" "c:\Windows\Syste...

csc.exe

system

17 Sep 2026 03:25...

cvtres.exe

cvtres.exe /NOLOGO /READONLY /MACHINE:IX86 "/OUT:C:\Windows\SystemTemp\RESC609.tmp" "c:\Windows\Syste...

csc.exe

system

18 Sep 2026 00:01...

conhost.exe

conhost.exe 0xffffffff -ForceV1

powershell.exe

system

21 Sep 2026 20:38...

conhost.exe

conhost.exe 0xffffffff -ForceV1

sc.exe

himds

Was the activity successful?

- ☐ Yes
- ☐ No
- ☒ Unknown

Verdict

- ☐ Malicious
- ☐ Non-malicious
- ☒ Suspicious — requires further investigation

Justification:

This registry change is a credential-theft staging step . It comes from three specific things in the log. First, the exact key and value touched, `WDigest\UseLogonCredential` set to `1`, is a well-documented Windows setting whose sole function is controlling whether WDigest caches plaintext credentials in LSASS; there's no other legitimate reason on a modern server to enable it, since WDigest itself is a legacy protocol Windows keeps only for backward compatibility. Second, the value changed from `<empty>` (unset, i.e. disabled by default on current Windows versions) to `1`, which is the specific direction that turns plaintext caching on rather than off — a change from `1` back to empty would read as remediation, not staging. Third, the process making the change, PID 4196, is the same PowerShell process that had just run a decoded, non-interactive script (the Ansible exec-wrapper) rather than a user manually opening Group Policy or the registry editor, meaning this was pushed automatically as part of a scripted sequence rather than a routine admin task. On the MITRE ATT&CK framework, it is an example of credential dumpingL LSASS Memory under modify registry.

3. Investigation Steps

1. Looked the process logs and alert logs
2. reearched about WDigest configuration change on google and Gemmni
3. KQL commands
4. MITTRE ATTACK research

Queries Used

```
// Paste queries here
```

```
DeviceProcessEvents  
| where DeviceName has "SOCSIM-SRV01"  
| where InitiatingProcessId == 4196  
| project Timestamp, FileName, ProcessCommandLine,  
InitiatingProcessFileName, AccountName  
| sort by Timestamp asc
```

```
DeviceRegistryEvents | where DeviceName has "SOCSIM-SRV01" | where Timestamp  
between (datetime(2026-08-25T20:40:29Z) .. datetime(2026-08-25T21:00:29Z)) |  
project Timestamp, ActionType, RegistryKey, RegistryValueName,  
RegistryValueData, InitiatingProcessFileName | sort by Timestamp asc
```

4. Categorisation

- ☐ ☒ **True Positive** — malicious activity confirmed
 - ☒ ☐ **Benign True Positive** — alert fired correctly, activity is legitimate
 - ☐ ☒ **False Positive** — alert fired incorrectly, rule needs review
-

5. Recommended Actions

Escalation:

- ☐ Urgent — phone call required
- ☒ Non-urgent — email sufficient
- ☐ No escalation required

Containment / Next Steps:

- Revert the registry change of `UseLogonCredential` back to `0` to encrypt all password logins.
- Capture a memory image of LSASS before rebooting
- Isolate the host from the network for further investigation
- Forced a password reset on any accounts logged on to `S0CSIM-SRV01`
- Restricting the registry key itself - Set explicit ACLs (permissions) on `HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\WDigest` so that only SYSTEM and a admins can change these registry by using GPO-enforced rule to `Disabled`

Alert Tuning:

- ☐ Tune recommended —
 - ☒ No tuning required
-

6. Lessons Learned

- Outside the alert, we have no real evidence of what happened, so sometimes calling in other experts and opinions to expand the analysis would be well.
 - An extra few days of analysis may make it better.
 - Further investigations could've been done.
 - Windows Defender decodes base64 already so no need for 3rd party tools to decode it anymore.
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►  About this template